

Chapter-4 Discrete Fourier Transform

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DFT is very powerful tool for frequency analysis of discrete time signals. DFT is itself a sequence rather than a function of a continuous variable and it corresponds to samples, equally spaced in frequency, of the Fourier transform of the signal. So DFT is a specific kind of discrete transform used in Fourier analysis.

DFT is used for transforming $x(n)$ of finite length into $X(k)$ of finite length.

Application of DFT:

1. statistical signal processing
2. frequency response of system
3. Convolution viz freg. domain.

Forward and Reverse Transform:

The DFT of discrete time signal $x(n)$ is denoted by $X(k)$ and defined by

$$\text{DFT: } X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}, \quad k=0, 1, \dots, N-1$$

Since, summation is taken for N -points, so it is called as N -point DFT.

We can obtain $x(n)$ from its DFT, called Inverse DFT,

given by

$$\text{IDFT: } x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi kn/N}, \quad n=0, 1, \dots, N-1$$

using twiddle factor;

$$\text{DFT: } X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}, \quad n=0,1,\dots,N-1$$

$$\text{IDFT: } x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn}, \quad n=0,1,\dots,N-1$$

where, $W_N = e^{j2\pi/N}$, called twiddle factor which makes the computation of DFT a bit easy and fast.

DFT as linear transformation:

$$\text{DFT: } X_N = W_N \cdot x_N$$

$$\text{IDFT: } x_N = W_N^{-1} X_N = \frac{1}{N} W_N^* X_N \quad \left(\because W_N^{-1} = \frac{1}{N} W_N^* \right)$$

where,

$$x_N = \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}, \quad X_N = \begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix}$$

$$W_N = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N & W_N^2 & \dots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{(N-1)} & W_N^{2(N-1)} & \dots & W_N^{(N-1)(N-1)} \end{bmatrix}$$

Relationship of DFT to others :

1. DFT to Fourier series coeff. of periodic signals.

$$\text{IDFT : } x_p(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}, \quad -\infty < n < \infty$$

where

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p(n) e^{-j2\pi kn/N}, \quad k=0, 1, \dots, N-1$$

$$\text{DFT : } X(k) = N c_k$$

2. DFT to Fourier transform of an aperiodic sequence

$$\text{DFT : } X(k) = X(\omega) \Big|_{\omega=2\pi k/N} = \sum_{n=-\infty}^{\infty} x(n) e^{j2\pi kn/N}, \quad k=0, 1, \dots, N-1$$

$$\text{IDFT : } x(n) = \hat{x}(n), \quad 0 \leq n \leq N-1$$

$$\text{where, } \hat{x}(n) = \begin{cases} x_p(n) = \sum_{l=-\infty}^{\infty} x(n-lN) & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

3. DFT to Z-transform

Let us consider a sequence $x(n)$ having the Z-transform.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \quad \text{ROC: include unit circle} \quad (1)$$

If $x(z)$ is sampled at the N equally spaced points on the unit circle $z_k = e^{j2\pi kn/N}$, $0, 1, 2, \dots, N-1$.

we obtain,

$$X(k) = X(z) \Big|_{z=e^{j2\pi kn/N}} = \sum_{n=-\infty}^{\infty} x(n) e^{-j2\pi kn/N}, \quad k=0,1,\dots,N-1$$

eqⁿ is identical to Fourier transform $X(\omega)$ evaluated at the N equally spaced frequencies $\omega_k = 2\pi k/N, k=0,1,\dots,N-1$

If the sequence $x(n)$ has a finite duration of length N its z -transform is

$$X(z) = \sum_{n=0}^{N-1} x(n) z^{-n}$$

Substituting IDFT relationship for $x(n)$, we have

$$X(z) = \sum_{n=0}^{N-1} \left[\frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi kn/N} \right] \cdot z^{-n}$$

$$X(z) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) \sum_{n=0}^{N-1} \left(e^{j2\pi k/N} z^{-1} \right)^n$$

$$X(z) = \frac{1-z^N}{N} \sum_{k=0}^{N-1} \frac{X(k)}{1 - e^{j2\pi k/N} z^{-1}}$$

This is identical to the frequency sampling form.

now, when evaluated over an unit circle,

$$X(\omega) = X(e^{j\omega}) = \frac{1 - e^{j\omega N}}{N} \sum_{k=0}^{N-1} \frac{X(k)}{1 - e^{-j(\omega - 2\pi k/N)}}$$

Properties of DFT:

1. Periodicity:-

If $x(n)$ and $X(k)$ are an N -point DFT pair, then

$$x(n+N) = x(n) \quad \text{for all } n$$

$$X(k+N) = X(k) \quad \text{for all } k.$$

2. Linearity:

$$\text{If } x_1(n) \xrightarrow[N]{\text{DFT}} X_1(k)$$

$$\text{and } x_2(n) \xrightarrow[N]{\text{DFT}} X_2(k) \text{ then,}$$

$$a_1 x_1(n) + a_2 x_2(n) \xrightarrow[N]{\text{DFT}} a_1 X_1(k) + a_2 X_2(k)$$

3. Symmetry:

If $x(n)$ and $X(k)$ are complex valued sequence i.e

$$x(n) = x_R(n) + j x_I(n), \quad 0 \leq n \leq N-1$$

$$X(k) = X_R(k) + j X_I(k), \quad 0 \leq k \leq N-1$$

i) If $x(n)$ is real valued,

$$X(N-k) = X^*(k) = X(-k)$$

ii) If $x(n)$ is real and even

$$x(n) = x(N-n) \quad 0 \leq n \leq N-1$$

iii) When $x(n)$ is real and odd

$$x(n) = -x(N-n), \quad 0 \leq n \leq N-1$$

iv) When $x(n)$ is purely imaginary

$$x(n) = j x_I(n).$$

4. Multiplication of two DFTs and Circular Convolution:
 The multiplication of two DFTs is equivalent to the circular convolution of their sequences in time domain.
 Mathematically,

$$\text{If } x_1(n) \xleftrightarrow[N]{\text{DFT}} X_1(K)$$

$$\text{and } x_2(n) \xleftrightarrow[N]{\text{DFT}} X_2(K) \text{ then}$$

$$x_1(n) \otimes x_2(n) \xleftrightarrow[N]{\text{DFT}} X_1(K) \cdot X_2(K),$$

let the circular convolution of $x_1(n)$ and $x_2(n)$ be $y(m)$,
 then

$$y(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2(m-n)_N \quad m=0, 1, \dots, N-1$$

Proof:

Let us consider two discrete sequences $x_1(n)$ and $x_2(n)$ of length N , then respective N -point DFTs are

$$X_1(K) = \sum_{n=0}^{N-1} x_1(n) e^{-j2\pi Kn/N} \quad L(1) \quad K=0, 1, \dots, N-1$$

$$\text{2nd, } X_2(K) = \sum_{l=0}^{N-1} x_2(l) e^{-j2\pi Kl/N} \quad L(II) \quad K=0, 1, \dots, N-1$$

multiplying eqn (I) & (II) we get, the result say $Y(K)$,

$$Y(K) = X_1(K) \cdot X_2(K) \quad K=0, 1, \dots, N-1$$

The IDFT of $Y(K)$ is

$$y(m) = \frac{1}{N} \sum_{k=0}^{N-1} Y(k) e^{j2\pi kn/N}$$

$$y(m) = \frac{1}{N} \sum_{k=0}^{N-1} x_1(k) \cdot x_2(k) e^{j2\pi kn/N}$$

Substituting the values of $x_1(k)$ & $x_2(k)$. we get

$$y(m) = \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{n=0}^{N-1} x_1(n) e^{j2\pi kn/N} \right] \left[\sum_{l=0}^{N-1} x_2(l) e^{j2\pi kl/N} \right] e^{j2\pi km/N}$$

Rearranging the summation,

$$y(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \left[\sum_{k=0}^{N-1} e^{j2\pi k(m-n-l)/N} \right] \quad \text{--- (iii)}$$

The inner sum in the bracket has the form

$$\sum_{k=0}^{N-1} a^k = \begin{cases} N & , a = 1 \\ \frac{1-a^N}{1-a} & , a \neq 1 \end{cases} \quad \text{--- (iv)}$$

where,

$$a = e^{j2\pi(m-n-l)/N}$$

we observe that $a=1$ when $m-n-l$ is multiple of N and $a^N=1$ for any value of $a \neq 0$.

then, the eqⁿ (iv) becomes

$$\sum_{k=0}^{N-1} a^k = \begin{cases} N, & l = m - n + pN = (m - n)_N, \quad p \text{ an integer} \\ 0, & \text{otherwise} \end{cases} \quad \text{--- (v)}$$

if we substitute the eqⁿ (v) in eqⁿ (iii), we get

$$y(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2((m-n)_N), \quad m = 0, 1, \dots, N-1$$

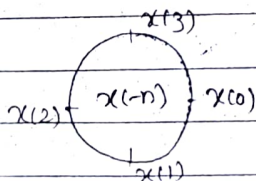
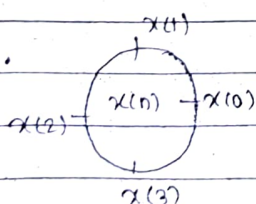
Here, the term $(m-n)_N$ is called circular convolution.

Thus, we conclude that multiplication of the DFTs of two sequences is equivalent to the circular convolution of two sequences in the time domain.

5. Time reversal :

if $x(n) \xrightarrow[N]{\text{DFT}} X(k)$ then

$$x((-n))_N = X(N-n) \xrightarrow[N]{\text{DFT}} X((-k))_N = X(N-k)$$



Hence, reversing the N-point sequence in time is equivalent to reversing the DFT values.

6. Circular time Shift:

If $x(n) \xrightarrow[N]{\text{DFT}} X(k)$ then

$$x((n-l))_N \xrightarrow[N]{\text{DFT}} X(k) e^{-j2\pi kl/N}$$

Shifting the sequence in time domain by l samples is equivalent to multiplying the sequence in freq. domain by $e^{-j2\pi kl/N}$ or w_N^{kl} .

7. Circular frequency Shift

If $x(n) \xrightarrow[N]{\text{DFT}} X(k)$ then

$$x(n) e^{j2\pi kn/N} \xrightarrow[N]{\text{DFT}} X((k-l))_N$$

$$\text{or } x(n) e^{-j2\pi kn/N} \xrightarrow[N]{\text{DFT}} X((k+l))_N$$

Multiplication of sequence $x(n)$ with the complex exponential sequence $e^{j2\pi kn/N}$ is equivalent to the circular shift of the DFT by l units in freq.

8. Multiplication of two sequences:

1st $x_1(n) \xrightarrow[N]{\text{DFT}} X_1(k)$

2nd

$x_2(n) \xrightarrow[N]{\text{DFT}} X_2(k)$ then

$$x_1(n) x_2(n) \xrightarrow[N]{\text{DFT}} \frac{1}{N} X_1(k) \otimes X_2(k)$$

Multiplication of two sequences in time domain equivalent to the circular convolution in frequency domain.

9. Circular Correlation.

$$\text{If } x(n) \xleftrightarrow[N]{\text{DFT}} X(K)$$

$$\text{and } y(n) \xleftrightarrow[N]{\text{DFT}} Y(K) \text{ then}$$

$$\tilde{r}_{xy}(l) \xleftrightarrow[N]{\text{DFT}} \tilde{R}_{xy}(K) = X(K) Y^*(K)$$

where

$$\tilde{r}_{xy}(l) = \sum_{n=0}^{N-1} x(n) y^*((n-l))_N$$

10. Parseval's Theorem:

for complex valued sequences $x(n)$ & $y(n)$,

$$\text{If } x(n) \xleftrightarrow[N]{\text{DFT}} X(K)$$

$$\text{and } y(n) \xleftrightarrow[N]{\text{DFT}} Y(K) \text{ then}$$

proof:

$$\sum_{n=0}^{N-1} x(n) y^*(n) = \frac{1}{N} \sum_{K=0}^{N-1} X(K) Y^*(K) \quad \text{--- (a)}$$

$$\text{we have, } \sum_{n=0}^{N-1} x(n) y^*(n) = \tilde{r}_{xy}(0)$$

$$\text{and } \tilde{r}_{xy}(l) = \frac{1}{N} \sum_{K=0}^{N-1} \tilde{R}_{xy}(K) e^{j2\pi K l / N}$$

$$\therefore \tilde{r}_{xy}(l) = \frac{1}{N} \sum_{K=0}^{N-1} X(K) Y^*(K) e^{j2\pi K l / N}$$

Hence, eqⁿ (a) follows by evaluating the IDFT at $l=0$.
If $y(n) = x(n)$, then,

$$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

Efficient Computation of DFT: FFT Algorithms.

Basically, the computational problem for the DFT is to compute the sequence $\{X(k)\}$ of N complex-valued numbers given another sequence of data $\{x(n)\}$ of length N , according to formula

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad 0 \leq k \leq N-1$$

where $W_N = e^{-j2\pi/N}$

Similarly, IDFT becomes

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-nk}, \quad 0 \leq n \leq N-1$$

We observe that, for each value of k , direct computation of $X(k)$ involves N complex multiplication ($4N$ real multiplication) and $N-1$ complex addition ($4N-2$ real additions). Consequently to compute all N values of the DFT requires N^2 complex multiplication and $N^2 - N$ complex additions.

Direct computation of the DFT is basically inefficient, primarily because it does not exploit the symmetry and periodicity properties of phase factor W_N . In particular, these two properties are

Symmetry property: $w_N^{k+(N/2)} = -w_N^k$

periodicity property: $w_N^{k+N} = w_N^k$

The computationally efficient algorithms described in this section, known collectively as Fast Fourier Transform (FFT) algorithms, exploit these two basic properties of the phase factor.

Direct Computation of the DFT:-

We have the DFT for an N-point sequence $x(n)$ as,

$$X(K) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \quad (i) \quad , K=0,1,\dots,N-1$$

IDFT of $X(K)$ is

$$\text{or } x(n) = \sum_{k=0}^{N-1} X(k) \cdot e^{j2\pi kn/N} \quad , n=0,1,\dots,N-1$$

Suppose $x(n)$ and $X(K)$ both are complex valued sequence i.e.

$$x(n) = x_R(n) + jx_I(n) \quad \text{--- (ii)} \quad 0 \leq n \leq N-1$$

$$X(K) = X_R(K) + jX_I(K) \quad \text{--- (iii)} \quad 0 \leq k \leq N-1$$

Substituting eqⁿ (ii) in eqⁿ (i) we get

$$X(K) = \sum_{n=0}^{N-1} [x_R(n) + jx_I(n)] e^{-j2\pi kn/N}$$

$$X(K) = \sum_{n=0}^{N-1} [x_R(n) + jx_I(n)] \left[\cos \frac{2\pi kn}{N} - j \sin \frac{2\pi kn}{N} \right]$$

using Euler's identity: $e^{j\theta} = \cos\theta + j\sin\theta$

$$N=4$$

$$x(k) = \sum_{n=0}^3 x(n) w_4^{kn} = x(0) \cdot w_4^0 + x(1) w_4^k + x(2) w_4^{2k} + x(3) w_4^{3k}$$

Total no. of multiplication = N^2 complex multiplication.

∴ Total multiplication for 4-point = $4N^2$ complex multiplication.

Total multiplication (real) = $4(N-1)$ real addition.

$$x_R(k) + j x_I(k) = \sum_{n=0}^{N-1} \left[\frac{x_R(n) \cos \frac{2\pi kn}{N} + x_I(n) \sin \frac{2\pi kn}{N}}{N} \right]$$

$$-j \sum_{n=0}^{N-1} \left[\frac{x_R(n) \sin \frac{2\pi kn}{N} - x_I(n) \cos \frac{2\pi kn}{N}}{N} \right]$$

Comparing real and imaginary parts, we get

$$x_R(k) = \sum_{n=0}^{N-1} \left[\frac{x_R(n) \cos \frac{2\pi kn}{N} + x_I(n) \sin \frac{2\pi kn}{N}}{N} \right] \quad \text{--- (iv)}$$

$$j x_I(k) = - \sum_{n=0}^{N-1} \left[\frac{x_R(n) \sin \frac{2\pi kn}{N} - x_I(n) \cos \frac{2\pi kn}{N}}{N} \right] \quad \text{--- (v)}$$

From eqⁿ (iv) & (v), we see that, For each value of k , the direct computation of DFT of $x(n)$ i.e. $x(k)$ requires $2N$ multiplication as shown by eqⁿ (iv) and $2N$ ^{other} multiplication as shown by eqⁿ (v), altogether $x(k)$ require $4N$ real multiplications and $4(N-1)$ real additions.

Since, $x(k)$ is computed for N different values of k , the direct computation of DFT of $x(n)$ requires $4N \times N = 4N^2$ real multiplications and $4(N-1) \times N = 4N(N-1)$ real additions. In other words we say that direct computation of DFT requires N^2 complex multiplication and $N(N-1)$ complex additions.

Thus, the direct computation of eqⁿ (iv) & (v), requires.

- i) $2N^2$ evaluations of trigonometric functions
- ii) $4N^2$ real multiplication or N^2 Complex multiplication
- iii) $4N(N-1)$ real addition or $N(N-1)$ complex addition.
- iv) A no. of indexing and addressing operations.

Applications of FFT algorithms:

1. The decomposition of an N -point DFT into successively smaller DFTs, leads to family of computationally efficient algorithms known collectively as FFT algorithms.
2. Linear filtering,
3. Correlation (cross correlation of two sequences).
4. Spectrum analysis

Comparison of DTFT and DFT

For the discrete time Fourier transform (DTFT) of aperiodic signals, no similar duality (between time domain and freq. domain) exists, since aperiodic signals and their Fourier transform are very different kinds of functions.

We know, DTFT is given by,

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \quad , \quad -\pi < \omega < \pi \quad \text{--- (i)}$$

Discrete Fourier transform (DFT) is obtained by sampling one cycle of Fourier transform.

DFT of $x(n)$ is given by

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \quad \text{--- (ii)}$$

Comparing eqn (i) & (ii), we obtain DFT from DTFT by substituting $\omega = \frac{2\pi k}{N}$.

Hence, $X(k) = X(\omega) \Big|_{\omega = \frac{2\pi k}{N}}$

- i) Continuous freq. spectrum $x(\omega)$ is replaced by discrete FT $X(k)$
- ii) Infinite summation in DTFT is replaced by finite summation in DFT.

Numericals:-

Q. Compute the DFT of the 4-point sequence $x(n) = \{0, 1, 2, 3\}$

Solⁿ: By Formula

Here, given, $x(n) = \{0, 1, 2, 3\}$ i.e. $N = 4$ ($\because$ 4-point seq.)

We know,

$$X(k) = \sum_{n=0}^{N-1} x(n) \cdot e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1$$

Since, ~~$N=4$~~ , It is a 4-point DFT, so, $N=4$,

$$\therefore X(k) = \sum_{n=0}^{4-1} x(n) \cdot e^{-j2\pi kn/4}, \quad k = 0, 1, \dots, 4-1$$

$$\therefore X(k) = \sum_{n=0}^3 x(n) e^{-j2\pi kn/4}, \quad k = 0, 1, \dots, 3$$

For $k=0$

$$X(0) = \sum_{n=0}^3 x(n) e^{-j2\pi (0) \cdot n/4}$$

$$= \sum_{n=0}^3 x(n) \cdot e^{-0}$$

$$= \sum_{n=0}^3 x(n)$$

$$= x(0) + x(1) + x(2) + x(3)$$

$$= 0 + 1 + 2 + 3$$

$$\therefore X(0) = 6$$

For $k=1$

$$X(1) = \sum_{n=0}^3 x(n) e^{-jn\pi/2}$$

$$= x(0) e^{-0} + x(1) \cdot e^{-j\pi/2} + x(2) \cdot e^{-j\pi} + x(3) \cdot e^{-j3\pi/2}$$

$$= 0 + 1(-j) + 2(-1) + 3(j)$$

$$\therefore X(1) = -2 + 2j$$

For $k=2$

$$X(2) = \sum_{n=0}^3 x(n) \cdot e^{-j2\pi n}$$

$$= x(0) e^{-0} + x(1) \cdot e^{-j2\pi} + x(2) \cdot e^{-j4\pi} + x(3) \cdot e^{-j6\pi}$$

$$= 0 + 1 \cdot (-1) + 2(1) + 3(-1)$$

$$\therefore X(2) = -2$$

For $k=3$

$$X(3) = \sum_{n=0}^3 x(n) e^{-j3\pi n/2}$$

$$= x(0) \cdot e^{-0} + x(1) \cdot e^{-j3\pi/2} + x(2) \cdot e^{-j3\pi} + x(3) \cdot e^{-j9\pi/2}$$

$$= 0 + 1(j) + 2(-1) + 3(-j)$$

$$\therefore X(3) = -2 - 2j$$

$$\therefore X(k) = \{ 0, -2 + 2j, -2, -2 - 2j \}$$

By Matrix method: i.e. using twiddle factor
 Since, it is a 4-point DFT, the DFT matrix is.

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_N^1 & W_N^2 & W_N^3 \\ 1 & W_N^2 & W_N^4 & W_N^6 \\ 1 & W_N^3 & W_N^6 & W_N^9 \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

where,

W_N is twiddle factor, value is $W_N = e^{-j2\pi/N}$

$$\therefore W_4 = e^{-j2\pi/4} = e^{-j\pi/2} = -j \quad W_4^5 = W_4^2 \cdot W_4^3 = -j$$

$$W_4^2 = (-j)^2 = -1 \quad W_4^6 = W_4^3 \cdot W_4^3 = j \cdot j = -1$$

$$W_4^3 = W_4 \cdot W_4^2 = (-j)(-1) = j \quad W_4^7 = W_4^3 \cdot W_4^4 = j$$

$$W_4^4 = W_4^2 \cdot W_4^2 = (-1)(-1) = 1 \quad W_4^8 = W_4^4 \cdot W_4^4 = 1 \cdot 1 = 1$$

$$W_4^9 = W_4^3 \cdot W_4^6 = -j$$

So,

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} p+1+q+3 \\ -j-2+3j \\ -1+2-3 \\ j-2-3j \end{bmatrix}$$

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 6 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix}$$

Q. Compute the 4-point DFT of $x(n) = \{1, 2, 3\}$
Find $x(10)$?

Soln:- Here, $x(n) = \{1, 2, 3\}$. Since it is 4-point DFT, so, $x(n)$ has 4-pt.
After zero padding, $x(n) = \{1, 2, 3, 0\}$

By matrix method:-

Since it is 4-point DFT, so, the DFT matrix is

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4 & W_4^2 & W_4^3 \\ 1 & W_4^2 & W_4^4 & W_4^6 \\ 1 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1+2+3 \\ 1-2j-3 \\ 1-2+3 \\ 1+2j-3 \end{bmatrix}$$

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 6 \\ -2-2j \\ 2 \\ -2+2j \end{bmatrix}$$

$$x(10) = 0, \quad x(100) = 0, \quad x(6) = 0.$$

Q compute the value of $x(n)$ from $x(k) = \{6, -2-2j, 2, -2+2j\}$

soln: Here, $x(k) = \{6, -2-2j, 2, -2+2j\}$

we can obtain $x(n)$ by using

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(k) W_N^{-KN}, \quad W_N = e^{-j2\pi/N}$$

OR we can obtain $x(n)$ by using (for 4-point IDFT)

$x(0)$		1	1	...	1	$x(0)$
$x(1)$					$W_N^{(N-1)k}$	$x(1)$
$x(2)$	$= \frac{1}{N}$	1	$W_N^{1 \times 2}$	...	$W_N^{2(N-1)k}$	1
$x(3)$		1	$W_N^{2 \times 2}$	...	$W_N^{3(N-1)k}$	1
$x(N-1)$		1	$W_N^{(N-1)k}$	...	$W_N^{(N-1)(N-1)k}$	$x(N-1)$

for 4-point IDFT

$x(0)$		1	1	1	1	$x(0)$
$x(1)$	$= \frac{1}{4}$	1	$j W_N^{1 \times 2}$	$+ W_N^{2 \times 2}$	$+ j W_N^{3 \times 2}$	$x(1)$
$x(2)$		1	$+ W_N^{2 \times 2}$	$W_N^{4 \times 2}$	$+ W_N^{6 \times 2}$	$x(2)$
$x(3)$		1	$+ j W_N^{3 \times 2}$	$+ W_N^{6 \times 2}$	$+ j W_N^{9 \times 2}$	$x(3)$

$$= \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 6 \\ -2-2j \\ 2 \\ -2+2j \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$$

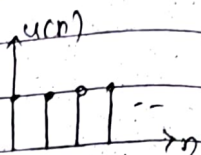
2013-fall
3.b) Q. Find the 4-point DFT of the following sequence:

$$x(n) = 2 \{ u(n) + u(n-1) - u(n-3) - u(n-4) \}$$

Solⁿ: For $n=0$

$$x(0) = 2 \{ u(0) + u(0-1) - u(0-3) - u(0-4) \}$$

$$= 2$$



for $n=1$

$$x(1) = 2 \{ u(1) + u(1-1) - u(1-3) - u(1-4) \}$$

$$= 2 \{ 1+1 \}$$

$$= 4$$

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

for $n=2$

$$x(2) = 2 \{ u(2) + u(2-1) - u(2-3) - u(2-4) \}$$

$$\therefore x(2) = 2(1+1) = 4$$

for $n=3$

$$x(3) = 2 \{ u(3) + u(3-1) - u(3-3) - u(3-4) \}$$

$$= 2(1+1-1)$$

$$= 2$$

$$\therefore x(n) = \{ 2, 4, 4, 2 \}, \quad N=4$$

Since, it is 4-point DFT, then, its DFT matrix is

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix}$$

$X(0)$	1	1	1	1	9	$2+4+4+2$
$X(1)$	1	$-j$	-1	j	4	$2-4j-4+2j$
$X(2)$	1	-1	1	-1	4	$2-4+4-2$
$X(3)$	1	j	-1	$-j$	2	$2+4j-4-2j$

$$\therefore \begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 12 \\ -2-2j \\ 0 \\ -2+2j \end{bmatrix}$$

Q. obtain the Circular Convolution of the following sequence
 $x_1 = \{1, 2, 3, 1\}$, $x_2 = \{4, 3, 2, 2\}$

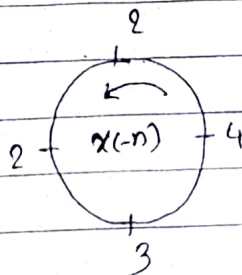
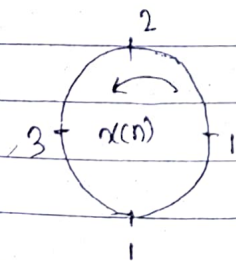
Solⁿ: Here, $x_1 = \{1, 2, 3, 1\}$ & $x_2 = \{4, 3, 2, 2\}$, $N=4$

We know,

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2((m-n)_N)$$

For $m=0$

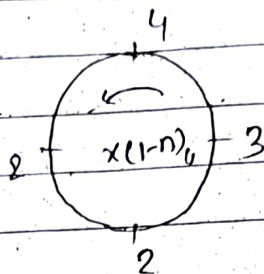
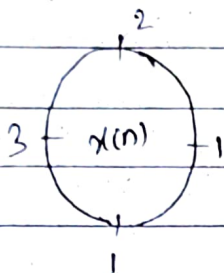
$$x_3(0) = \sum_{n=0}^3 x_1(n) \cdot x_2((-n)_4) \quad \because N=4$$



$$x_3(0) = 3 \times 2 + 2 \times 2 + 1 \times 4 + 1 \times 3 = 17$$

For $m=1$

$$x_3(1) = \sum_{n=0}^3 x_1(n) \cdot x_2(1-n)_4$$

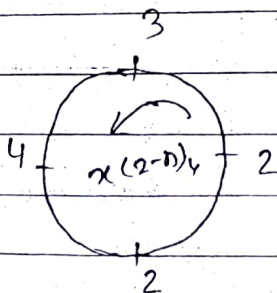
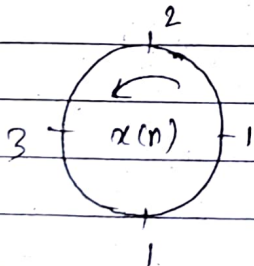


$$x_3(1) = 2 \times 4 + 1 \times 3 + 1 \times 2 + 3 \times 2$$

$$= 19$$

For $m=2$

$$x_3(2) = \sum_{n=0}^3 x_1(n) \cdot x_2(2-n)_4$$

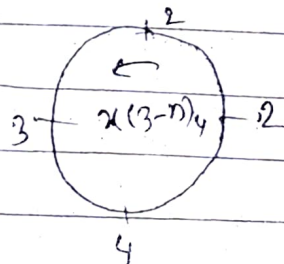
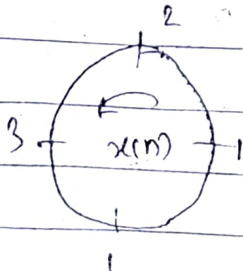


$$x_3(2) = 2 \times 3 + 1 \times 2 + 1 \times 2 + 3 \times 4$$

$$= 22$$

For $m=3$

$$x_3(3) = \sum_{n=0}^3 x_1(n) \cdot x_2(3-n)_4$$



$$x_3(3) = 2 \times 2 + 1 \times 2 + 1 \times 4 + 3 \times 3 = 19$$

$$\therefore x_3(m) = \{17, 19, 22, 19\}$$

According to linear Convolution i.e

	1	2	3	1
4	4	8	12	4
3	3	6	9	3
2	2	4	6	2
2	2	4	6	2

OR

	4	3	2	2
1	4	3	2	2
2	8	6	4	4
3	12	9	6	6
1	4	3	2	2

$$x_5(m) = \{4, 11, 20, 19, 13, 8, 2\}$$

$$x_2(m) = \{4, 11, 20, 19, 13, 8, 2\}$$

2013 Spring
3.1)

What is padding? find linear convolution through circular convolution with padding of zeros for the following sequences.

$$x(n) = \{1, 2, 4\}, \quad h(n) = \{1, 2\}$$

Soln:-

Zero padding:

The process of adding zeros to the original sequence for the efficient computation of DFT is called zero padding. It does not provide any additional information about the spectrum $x(j\omega)$ of sequence $x(n)$.

Here, given, $x(n) = \{1, 2, 4\}$ $h(n) = \{1, 2\}$, $M=2$

i.e $L=3$ $\therefore$ Total length of sequence $= 3+2-1=4$

After zero padding

i.e $N=L+M-1=3+2-1=4$

$$x(n) = \{1, 2, 4, 0\} \quad h(n) = \{1, 2, 0, 0\}$$

We know,

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2(m-n) \quad , m=0, 1, \dots, N-1$$

Not use this formula

$\therefore$ we have to calculate 4-point DFT i.e. $N=4$

we have, $X(k) = W_N \cdot x_N$ i.e. $X(k) = W_4 \cdot x_4$

$$\therefore \begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 7 \\ -3-2j \\ 2 \\ -3+2j \end{bmatrix}$$

$$\therefore X(k) = \{ 7, -3-2j, 2, -3+2j \}$$

Let us find $H(k)$:

we have, $H(k) = W_N \cdot h_N$ i.e. $H(k) = W_4 \cdot h_4$

$$\begin{bmatrix} H(0) \\ H(1) \\ H(2) \\ H(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} h(0) \\ h(1) \\ h(2) \\ h(3) \end{bmatrix}$$

$H(0)$		1	1	1	1		1		3
$H(1)$	=	1	-j	-1	+j		2		$1-2j$
$H(2)$		1	-1	1	-1		0		-1
$H(3)$		1	j	-1	j		0		$1+2j$

$$\therefore H(k) = \{3, 1-2j, -1, 1+2j\}$$

now,

$$Y(k) = X(k) \cdot H(k)$$

$$= \{7, -3-2j, 2, -3+2j\} \cdot \{3, 1-2j, -1, 1+2j\}$$

$$Y(0) = X(0) \cdot H(0) = 7 \times 3 = 21$$

$$Y(1) = X(1) \cdot H(1) = (-3-2j) \times (1-2j) = -7+j4$$

$$Y(2) = X(2) \cdot H(2) = 2 \times (-1) = -2$$

$$Y(3) = X(3) \cdot H(3) = (-3+2j) \times (1+2j) = -7-j4$$

$$\therefore Y(k) = \{Y(0), Y(1), Y(2), Y(3)\}$$

$$\therefore Y(k) = \{21, -7+j4, -2, -7-j4\} \quad \underline{\underline{R}}$$

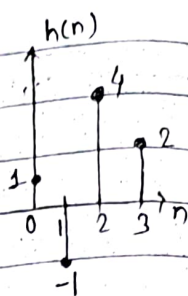
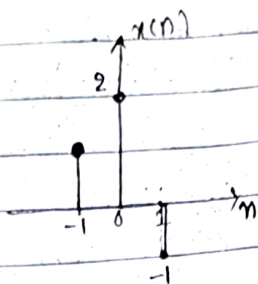
2015 Fall
Q. 4.b) Suppose $x(n) = \{1, 2, -1\}$ be the input to an LTI system with impulse response $h(n) = \{1, -1, 4, 2\}$. perform circular convolution to calculate response of the system, $y(n)$.

Solⁿ:- Here, given, $x(n) = \{1, 2, -1\}$ i.e. $L=3$

$h(n) = \{1, -1, 4, 2\}$ i.e. $M=4$

$\therefore$ DFT point $N = L+M-1 = 3+4-1 = 6$
i.e. Range is $(-1 \text{ to } 4)$

For zero padding
 sequence range $-N$
 $(-1+0)$ to $(1+3)$
 -1 to 4



$$x(n) = \{1, 2, -1, 0, 0, 0\}$$

$$h(n) = \{0, 1, -1, 4, 2, 0\}$$

we know, circular convolution of $x_1(n)$ & $x_2(n)$ is

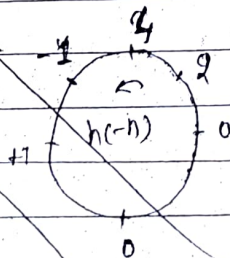
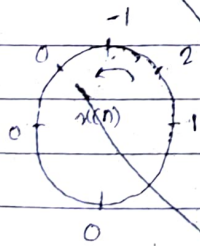
$$y(m) = x_2(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2(m-n)_N$$

Similarly

$$y(m) = \sum_{n=0}^{N-1} x(n) \cdot h(m-n)_N$$

for $m=-1$,

$$y(-1) = \sum_{n=0}^4 x(n) \cdot h(-n)_N$$



$$y(-1) = -1 \times 2 + 2 \times 0 + 1 \times 0 + 0 \times 1 + 0 \times (-1) + 0 \times 4$$

$$= -2$$

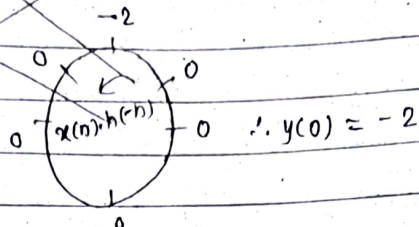
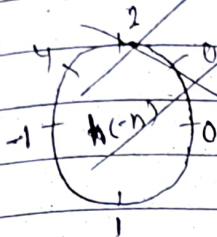
Linear Convolution

	1	-1	4	2
1	1	-1	4	2
2	2	-2	8	4
-1	-1	1	-4	-2

$\therefore y(n) = \{1, 2, 1, 11, 0, -2\}$

for $m=0$

$$y(0) = \sum_{n=-1}^4 x(n) \cdot h(-n)$$



Q2. Circular Convolution by using matrix method,

i.e.

$y(0)$	$h(0)$	$h(N-1)$	$h(N-2)$	$\dots$	$h(2)$	$h(1)$	$x(0)$
$y(1)$	$h(1)$	$h(0)$	$h(N-1)$	$\dots$	$h(3)$	$h(2)$	$x(1)$
$y(2)$	$h(2)$	$h(1)$	$h(0)$	$\dots$	$h(4)$	$h(3)$	$x(2)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$y(N-1)$	$h(N-1)$	$h(N-2)$	$h(N-3)$	$\dots$	$h(1)$	$h(0)$	$x(N-1)$

$$\therefore \begin{bmatrix} y(-1) \\ y(0) \\ y(1) \\ y(2) \\ y(3) \\ y(4) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 2 & 4 & -1 \\ -1 & 1 & 0 & 0 & 2 & 4 \\ 4 & -1 & -1 & 0 & 0 & 2 \\ 2 & 4 & -1 & 1 & 0 & 0 \\ 0 & 2 & 4 & -1 & 1 & 0 \\ 0 & 0 & 2 & 4 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ -2 \end{bmatrix} \quad \text{i.e. } y(k) = \{1, 1, 1, 1, 0, -2\}$$

Q. 8. Find the circular convolution of the sequence
 $x_1(n) = \{0, 1, 2, 3\}$ $x_2(n) = \{2, 1, 1, 2\}$

By tabular method

Solⁿ we know,

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2(m-n)_N$$

	0	1	2	3
2	0	2	4	6
1	0	1	2	3
1	0	1	2	3
2	0	2	4	6

For $m=0$

$$x_3(0) = \sum_{n=0}^3 x_1(n) \cdot x_2(-n)_4$$

$$= 7$$

$$x_3(m) = \{0, 2, 5, 9, 7, 7, 6\}$$

For $m=1 \Rightarrow x_3(1) = 9$

For $m=2 \Rightarrow x_3(2) = 11$

For $m=3 \Rightarrow x_3(3) = 9$

Q. Find DFT of the sequence $x(n) = \{1, 2, 3, 4\}$ using DIT-FFFT
i.e. $N=4$

Soln:- using Decimation in time FFT:

no. of stage = $4 = 2^2 \rightarrow 2$ -stage

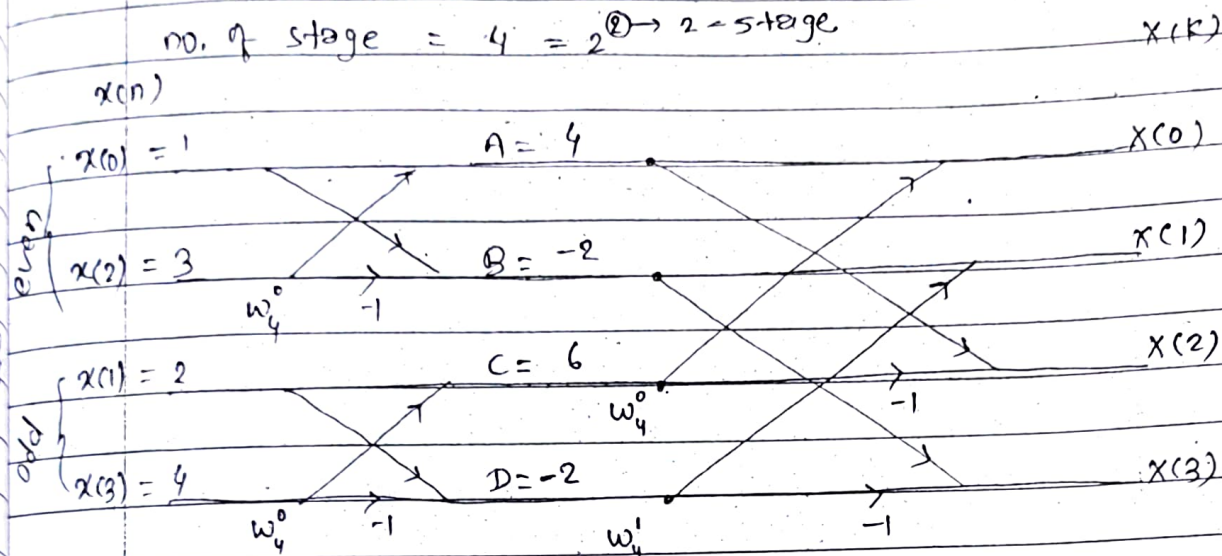


fig: 2-stage DIT-FFFT

1st stage calculation

$$A = 1 + 3 \cdot w_4^0 = 4$$

$$B = 3 \cdot w_4^1 (-1) + 1 = -2$$

$$C = 2 + 4 \cdot w_4^0 = 6$$

$$D = 4 \cdot w_4^1 (-1) + 2 = -2$$

2nd stage calculation

$$X(0) = 4 + 6 \cdot w_4^0 = 10$$

$$X(1) = -2 + (-2) \cdot w_4^1 = -2 + 2j$$

$$X(2) = 6 \cdot w_4^0 (-1) + 4 = -2$$

$$X(3) = -2 \cdot w_4^1 (-1) + (-2) = -2 + 2j$$

$$\therefore X(k) = \{10, -2+2j, -2, -2+2j\}$$

where, i.e. where, $w_4^1 = e^{-j2\pi/4} = e^{-j\pi/2}$

$$\therefore w_4^1 = \cos \frac{\pi}{2} - j \sin \frac{\pi}{2} = 0 - j(1) = -j$$

Q. Find DFT of $x(n) = \{1, 2, 3, 4\}$ using decimation in frequency FFT

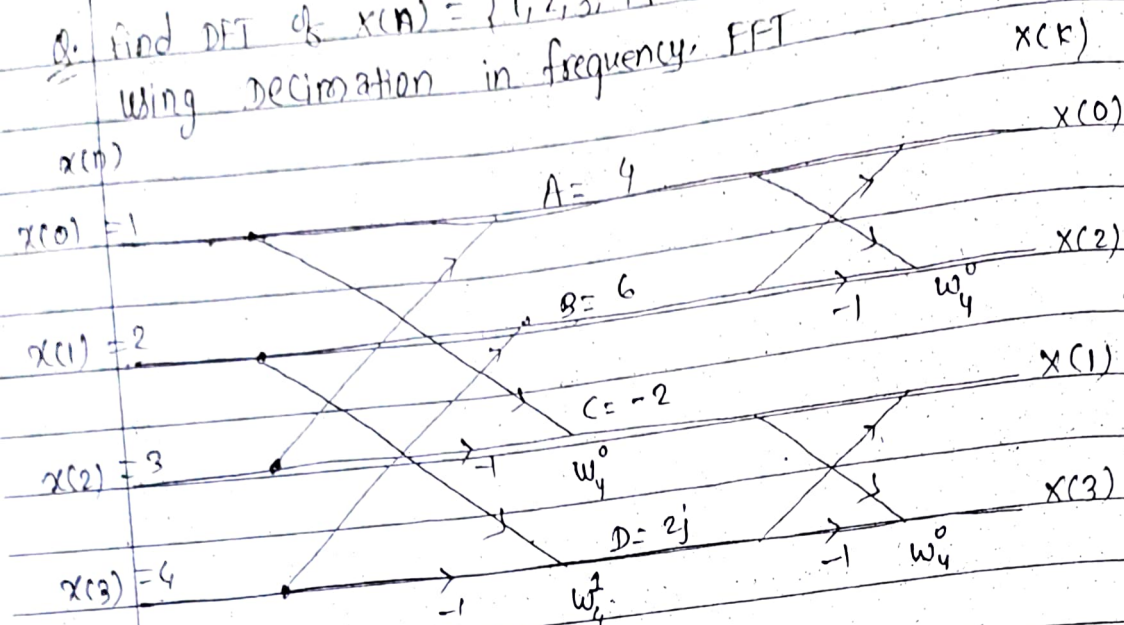


Fig: 2-stage DIF FFT

1st stage calculation

$$A = 1 + 3 = 4$$

$$B = 2 + 4 = 6$$

$$C = 3(-1) + 1 \cdot w_4^0 = -2$$

$$D = [4(-1) + 2] w_4^1 = 2j$$

2nd-stage calculation

$$X(0) = 4 + 6 = 10$$

$$X(2) = (6(-1) + 4) w_4^0 = -2$$

$$X(1) = -2 + 2j$$

$$X(3) = [2j(-1) + (-2)] w_4^0 = -2 - 2j$$

$$\therefore X(k) = \{10, -2 + 2j, -2, -2 - 2j\}$$

Q. Find DFT of the sequence $\{1, -2, 4, -9\}$ using decimation in frequency FFT

Soln: Let $x(n) = \{1, -2, 4, -9\}$

No. of stage = 2 $\Rightarrow$ 2-stage

Using Decimation in freq. FFT

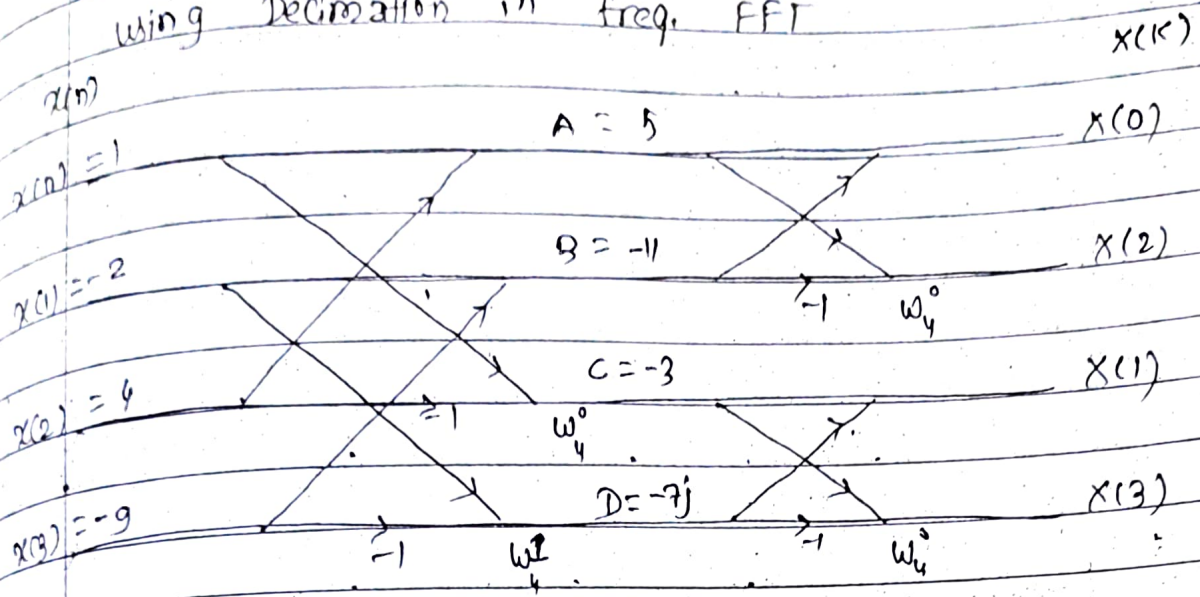


fig: 2-stage DIF FFT

1st stage calculation

$$A = 1 + 4 = 5$$

$$B = -2 + (-9) = -11$$

$$C = [4(-1) + 1]w_4^0 = -3$$

$$D = [-9(-1) + (-2)]w_4^1 = -7j$$

2nd stage calculation

$$X(0) = 5 + (-11) = -6$$

$$X(2) = [-11(-1) + 5]w_4^0 = 16$$

$$X(1) = -3 - 7j$$

$$X(3) = [-7j(-1) + (-3)]w_4^1 = -3 + 7j$$

$$\therefore X(k) = \{-6, -3 - 7j, 16, -3 + 7j\}$$

Q. What is the importance of FFT? Find 8-point DFT of sequence $x(n) = \{1, 2, 4, 5, 3\}$ using DIT-FFT algorithm

Here, $x(n) = \{1, 2, 4, 5, 3\}$

Since it is 8-point DFT, i.e. $N=8$, i.e. $x(n)$ must have the length of 8, i.e. using zero padding.

$$x(n) = \{1, 2, 4, 5, 3, 0, 0, 0\}$$

now,

using DIT-FFT algorithm

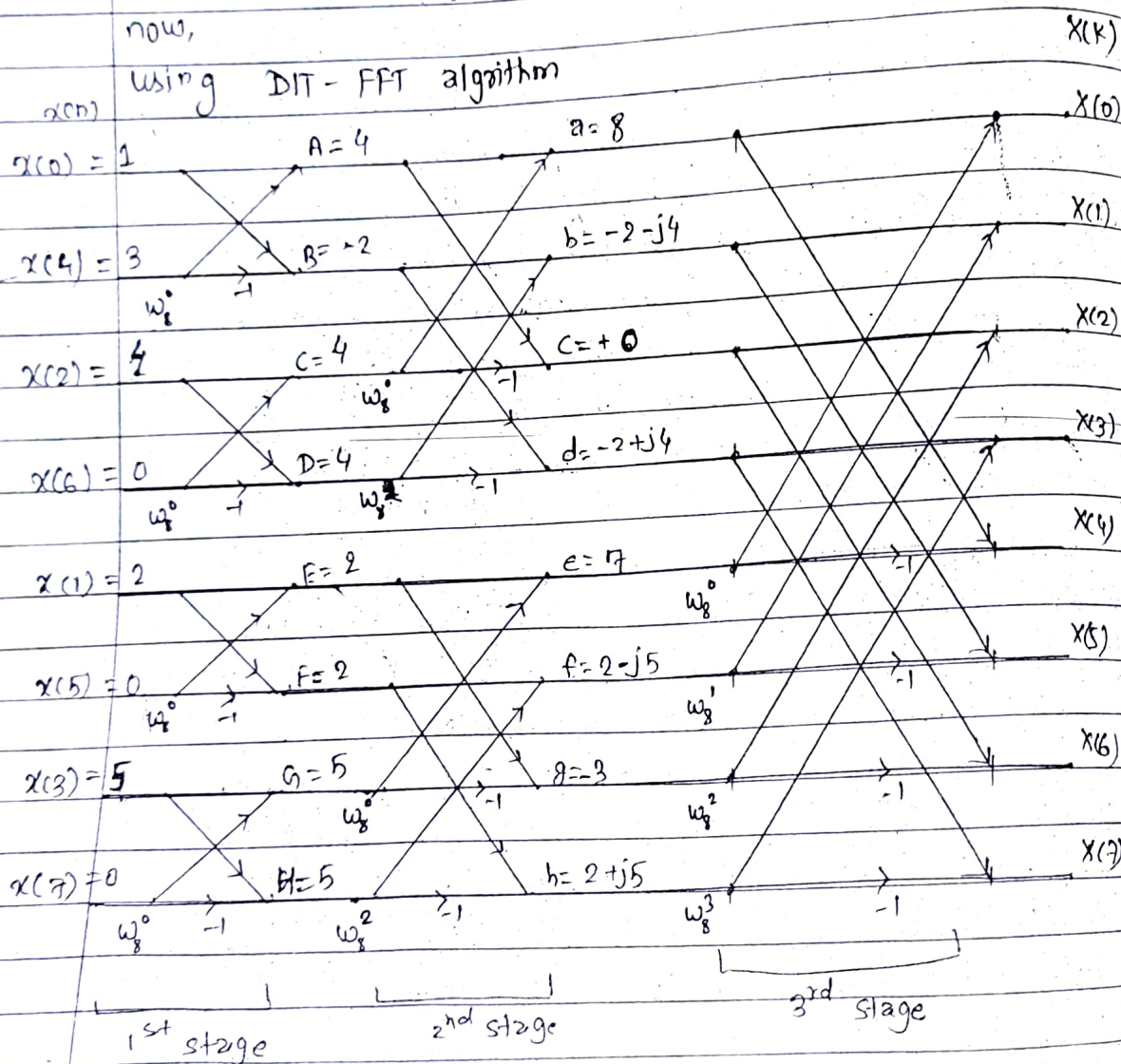


Fig: DIT - FFT (3-stage)

$$2^3 = 8$$

$$w_8^0 = 1, \quad w_8^1 = +0.707 - j0.707, \quad w_8^2 = (w_8^1)^2 = -j$$

$$\& w_8^3 = w_8^1 \cdot w_8^2 = -0.707 + j0.707$$

1st stage calculation

2nd stage calculation

$$A = 1 + 3 \cdot \omega_8^0 = 4$$

$$a = 4 + 4 \cdot \omega_8^0 = 8$$

$$B = 3 \cdot \omega_8^0 \cdot (-1) + 1 = -2$$

$$b = -2 + 4 \cdot \omega_8^2 = -2 + 4(-j) = -2 - j4$$

$$C = 4 + 0 \cdot \omega_8^0 = 4$$

$$c = 4 \cdot \omega_8^0 \cdot (-1) + (4) = 0$$

$$D = [0 \cdot \omega_8^0 \cdot (-1) + 4] = 4$$

$$d = 4 \cdot \omega_8^2 \cdot (-1) + (-2) = 4(-j)(-1) + (-2) = -2 + j4$$

$$E = 2 + 0 \cdot \omega_8^0 = 2$$

$$e = 2 + 5 \cdot \omega_8^0 = 7$$

$$F = 0 \cdot \omega_8^0 \cdot (-1) + 2 = 2$$

$$f = 2 + 5 \cdot \omega_8^2 = 2 + 5(-j) = 2 - j5$$

$$G = 5 + 0 \cdot \omega_8^0 = 5$$

$$g = 5 \cdot \omega_8^0 \cdot (-1) + 2 = -3$$

$$H = 0 \cdot \omega_8^0 \cdot (-1) + 5 = 5$$

$$h = 5 \cdot \omega_8^2 \cdot (-1) + 2 = 5(-j)(-1) + 2 = 2 + j5$$

3rd stage calculation.

$$X(0) = 8 + 7 \cdot \omega_8^0 = 15$$

$$X(1) = (-2 - j4) + (2 - j5) \cdot \omega_8^1 = -4.121 - j8.949$$

$$X(2) = 0 + 2(-3) \cdot \omega_8^2 = 3j$$

$$X(3) = (-2 + j4) + (2 + j5) \cdot \omega_8^3 = 0.121 - j0.949$$

$$X(4) = 7 \cdot \omega_8^0 \cdot (-1) + 8 = 1$$

$$X(5) = (2 - j5) \cdot \omega_8^1 \cdot (-1) + (-2 - j4) = 0.121 + j0.949$$

$$X(6) = -3 \cdot \omega_8^2 \cdot (-1) + 0 = -3j$$

$$X(7) = (2 + j5) \cdot \omega_8^3 \cdot (-1) + (-2 + j4) = -4.121 + j8.949$$

$$\therefore X(k) = \{ 15, -4.121 - j8.949, 3j, 0.121 - j0.949, 1, 0.121 + j0.949, -3j, -4.121 + j8.949 \}$$

Combination of conjugate pair

$$x(1) + x(7)$$

$$x(2) + x(6)$$

$$x(3) + x(5)$$

2015-Spring
4.81

Q. Find the 8-point DFT of $x(n)$,

$$x(n) = \sin \frac{3\pi n}{4}, \quad \text{for } 0 \leq n \leq 7.$$

Solⁿ:-

$$\text{Here, } x(n) = \sin \frac{3\pi n}{4} \quad 0 \leq n \leq 7$$

For $n=0$

$$x(0) = \sin \left(\frac{3\pi}{4} \times 0 \right) = 0$$

For $n=1$

$$x(1) = \sin \left(\frac{3\pi}{4} \right) = \frac{1}{\sqrt{2}} = 0.707$$

For $n=2$

$$x(2) = \sin \left(\frac{3\pi}{2} \right) = -1$$

For $n=3$

$$x(3) = \sin \left(\frac{9\pi}{4} \right) = 0.707$$

For $n=4$

$$x(4) = \sin(3\pi) = 0$$

For $n=5$

$$x(5) = \sin \left(\frac{15\pi}{4} \right) = -0.707$$

For $n=6$

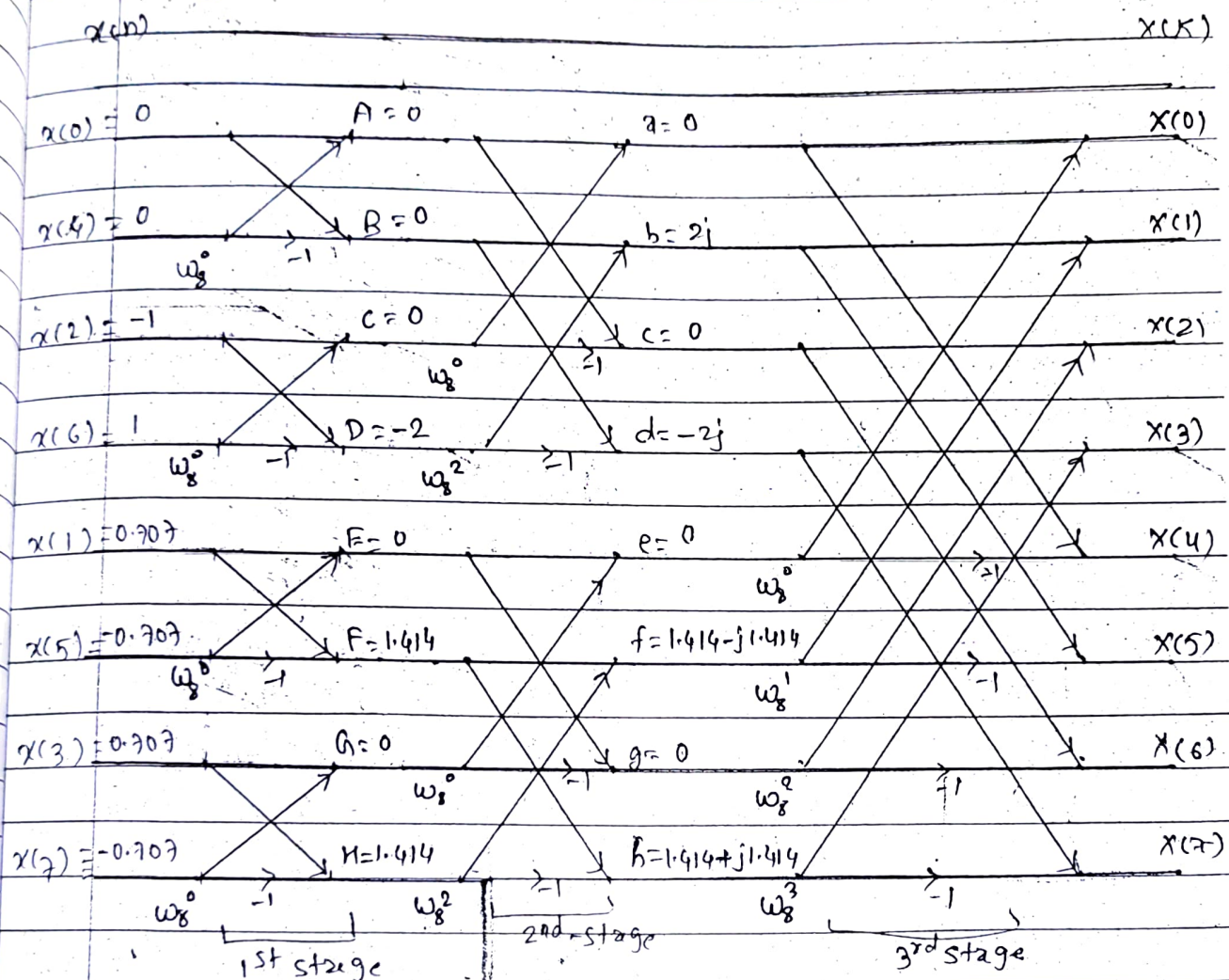
$$x(6) = \sin \left(\frac{9\pi}{2} \right) = 1$$

For $n=7$

$$x(7) = \sin \left(\frac{21\pi}{4} \right) = -0.707$$

$$\therefore x(n) = \{0, 0.707, -1, 0.707, 0, -0.707, 1, -0.707\}$$

using DIT FFT



1st stage calculation

$$A = 0 + 0 \cdot \omega_8^0 = 0$$

$$B = 0 \cdot \omega_8^0 \cdot (-1) + 0 = 0$$

$$C = -1 + 1 \cdot \omega_8^0 = 0$$

$$D = 1 \cdot \omega_8^0 \cdot (-1) - 1 = -2$$

$$E = 0.707 - 0.707 \cdot \omega_8^0 = 0$$

$$F = -0.707 \cdot \omega_8^0 \cdot (-1) + 0.707 = 1.414$$

$$G = 0.707 + (-0.707) \cdot \omega_8^0 = 0$$

$$H = -0.707 \cdot \omega_8^0 \cdot (-1) + 0.707 = 1.414$$

2nd stage calculation

$$a = 0 + 0 \cdot \omega_8^0 = 0$$

$$b = 0 + (-2) \cdot \omega_8^2 = -2j$$

$$c = 0 \cdot \omega_8^0 \cdot (-1) + 0 = 0$$

$$d = -2 \cdot \omega_8^2 \cdot (-1) + 0 = 2j$$

$$e = 0 + 0 \cdot \omega_8^0 = 0$$

$$f = 1.414 + 1.414 \cdot \omega_8^2 = 1.414 - j1.414$$

$$g = 0 \cdot \omega_8^0 \cdot (-1) + 0 = 0$$

$$h = 1.414 - 1.414 \cdot \omega_8^2 = 1.414 + j1.414$$

3rd stage calculation:

$$X(0) = 0 + 0 \cdot \omega_8^0 = 0$$

$$X(1) = 0j + (1.414 - j1.414) \omega_8^1 = j0.000604 - j0.000604$$

$$X(2) = 0 + 0 \cdot \omega_8^2 = 0$$

$$X(3) = -2j + (1.414 + j1.414) \omega_8^3 = -j3.999 = -j4$$

$$X(4) = 0 \cdot \omega_8^4(-1) + 0 = 0$$

$$X(5) = (1.414 - j1.414) \omega_8^{5(-1)} + 2j = j0.000604 = j4$$

$$X(6) = 0 \cdot \omega_8^6(-1) + 0 = 0$$

$$X(7) = (1.414 + j1.414) \omega_8^{7(-1)} + (-2j) = -j3.999 = -j4$$

$$\therefore x(k) = \{0, j0.000604, 0, -j4, 0, -j0.000604, 0, j4\}$$

$$x(k) = \{0, j0.000604, 0, -j4, 0, j4, 0, -j0.000604\} \quad \mathcal{Q}$$

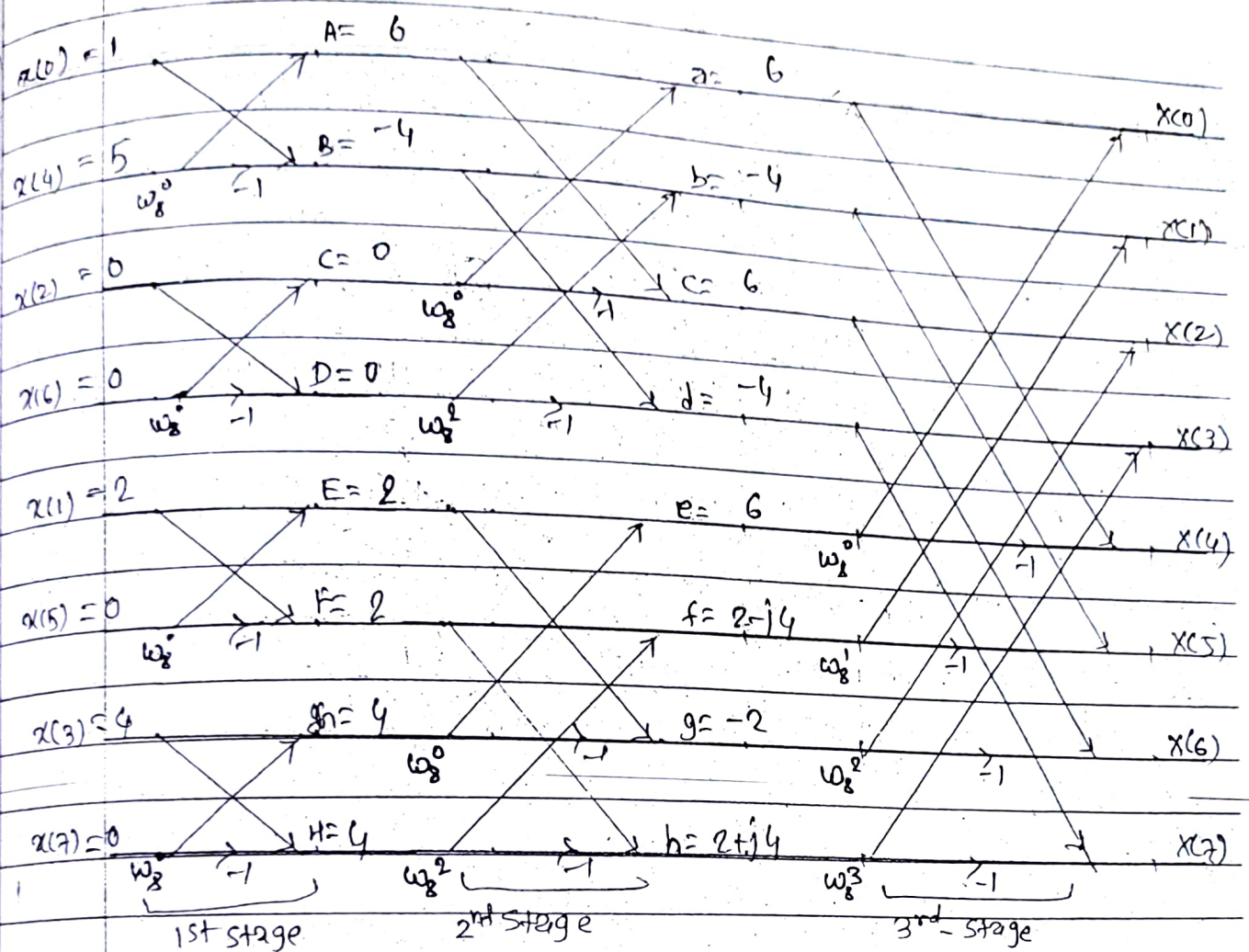
2015-16
4.27 = 8

Find the DFT of the signal $x(n) = \{1, 2, 0, 4, 5\}$ using radix-2 decimation in time algorithm.

Soln:- Here, given, $x(n) = \{1, 2, 0, 4, 5\}$

Since, radix-2 DIT is 8-point algorithm so, after zero padding,

$$x(n) = \{1, 2, 0, 4, 5, 0, 0, 0\}$$



1st stage calculation

2nd stage calculation

$$A = 1 + 5w_8^0 = 6$$

$$B = 5 \cdot w_8^0 \cdot (-1) + 1 = -4$$

$$C = 0$$

$$D = 0$$

$$E = 2 + 0 \cdot w_8^0 = 2$$

$$F = 0 + 2 = 2$$

$$G = 4 + 0 = 4$$

$$H = 0 + 4 = 4$$

$$a = 6 + 0 = 6$$

$$b = -4 + 0 = -4$$

$$c = 0 + 6 = 6$$

$$d = 0 + (-4) = -4$$

$$e = 2 + 4 \cdot w_8^0 = 2 + 4 = 6$$

$$f = 2 + 4w_8^1 = 2 + 4 \cdot w_8^1 = 2 - j4$$

$$g = 4 \cdot w_8^2 \cdot (-1) + 2 = -2$$

$$h = 4 \cdot w_8^3 \cdot (-1) + 2 = 2 + j4$$

3rd stage calculation

$$X(0) = 6 + 6 \cdot \omega_8^0 = 12$$

$$X(1) = -4 + (2 - j4) \omega_8^1 = -5.414 - j4.242$$

$$X(2) = 6 + (1 - 2) \omega_8^2 = 6 + 2j$$

$$X(3) = -4 + (2 + j4) \omega_8^3 = -8.242 + j4.414 = -2.586 - j4.242$$

$$X(4) = 6 \cdot \omega_8^4 (-1) + 6 = 0$$

$$X(5) = (2 - j4) (\omega_8^1 X(1)) + (-4) = -8.242 + j4.414 = -2.586 + j4.242$$

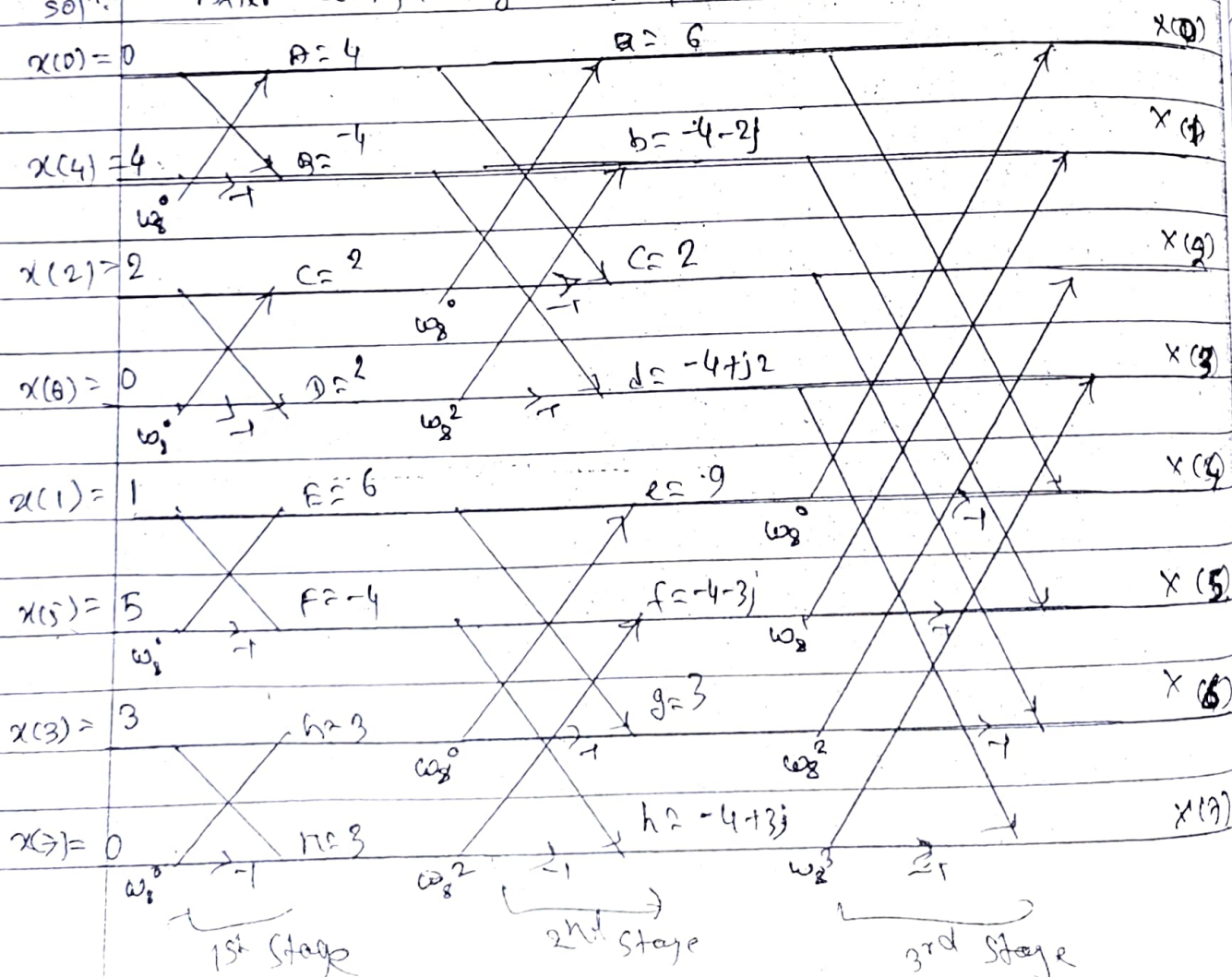
$$X(6) = (-2 \omega_8^2) (-1) + 6 = 6 - 2j$$

$$X(7) = (2 + j4) (\omega_8^3) (-1) + (-4) = -5.414 + j4.242$$

$$\therefore X(k) = \{12, -5.414 - j4.242, 6 + 2j, -2.586 - j4.242, 0, -2.586 + j4.242, 6 - 2j, -5.414 + j4.242\}$$

8. Find 8-point DFT of $x(n) = \{0, 1, 2, 3, 4, 5\}$, using DIT-FFT algorithm.

soln:- After zero padding $x(n) = \{0, 1, 2, 3, 4, 5, 0, 0\}$ $\because N=8$.



1st stage calculation

2nd stage calculation

$$A = 0 + 4 \cdot \omega_8^0 = 4$$

$$B = 4 \cdot \omega_8^0 \cdot (-1) + 0 = -4$$

$$C = 2 + 0 \cdot \omega_8^1 = 2$$

$$D = 0 \cdot \omega_8^1 \cdot (-1) + 2 = 2$$

$$E = 1 + 5 \cdot \omega_8^2 = 6$$

$$F = 5 \cdot \omega_8^2 \cdot (-1) + 1 = -4$$

$$G = 3 + 0 \cdot \omega_8^3 = 3$$

$$H = 0 \cdot \omega_8^3 \cdot (-1) + 3 = 3$$

$$a = 4 + 2 \omega_8^0 = 6$$

$$b = -4 + 2 \omega_8^1 = -4 - 2j$$

$$c = 2 \cdot \omega_8^1 \cdot (-1) + 4 = 2$$

$$d = 2 \omega_8^1 \cdot (-1) + (-4) = -4 + j2$$

$$e = 6 + 3 \cdot \omega_8^2 = 9$$

$$f = -4 + 3 \cdot \omega_8^2 = -4 - 3j$$

$$g = 3 \cdot \omega_8^3 \cdot (-1) + 6 = 3$$

$$h = 3 \cdot \omega_8^3 \cdot (-1) + (-4) = -4 + 3j$$

3rd stage calculation

$$x(0) = 6 + 9 \omega_8^0 = 15$$

$$x(1) = -4 - 2j + (-4 - 3j) \omega_8^1 = -8.949 - j1.293$$

$$x(2) = 2 + 3 \cdot \omega_8^2 = 2 + j3$$

$$x(3) = -4 + j2 + (-4 + 3j) \omega_8^1 = 0.949 + j2.707$$

$$x(4) = 9 \cdot \omega_8^0 \cdot (-1) + 6 = -3$$

$$x(5) = (-4 - 3j) \omega_8^1 \cdot (-1) + (-4 - 2j) = 0.949 - j2.707$$

$$x(6) = 3 \omega_8^2 \cdot (-1) + 2 = 2 + 3j$$

$$x(7) = (-4 + 3j) \omega_8^3 \cdot (-1) + (-4 + j2) = -8.949 + j1.293$$

$$\therefore x(k) = \{15, -8.949 - j1.293, 2 + j3, 0.949 + j2.707, -3, 0.949 - j2.707, 2 + 3j, -8.949 + j1.293\}$$

Find 8-point DFT of sequence $x(n) = [1, 2, 4, 5, 3]$ using DIT FFT algorithm

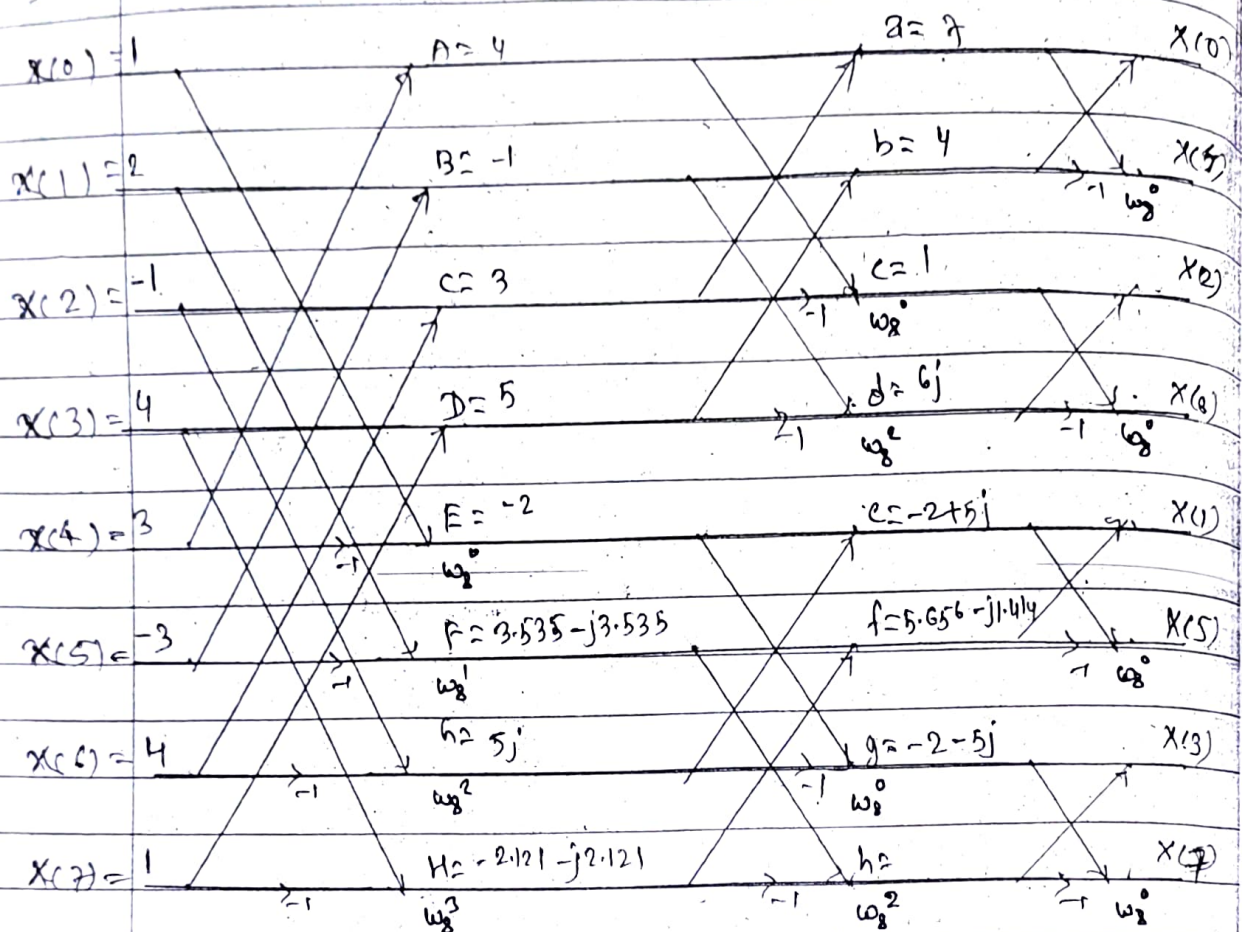
Soln: since, it is 8-point DFT i.e. $N=8$ so, after zero padding

$$x(n) = \{1, 2, 4, 5, 3, 0, 0, 0\}$$

20/11/2024
u.b

Q find 8-point DFT of the sequence $\{1, 2, -1, 4, 3, -3, 4, 1\}$ using radix-2 decimation in frequency domain

Soln: Here, $x(n) = \{1, 2, -1, 4, 3, -3, 4, 1\}$



1st stage calculation

2nd stage calculation

3rd stage calculation

$$A = 1 + 3 = 4$$

$$B = 2 + (-3) = -1$$

$$C = -1 + 4 = 3$$

$$D = 4 + 1 = 5$$

$$E = [3(-1) + 1] \omega_8^0 = -2$$

$$F = [3(-1) + 2] \omega_8^1 = 3.535 - j3.535$$

$$G = [4(-1) + (-1)] \omega_8^2 = 5j$$

$$H = [1(-1) + 4] \omega_8^3 = -2.121 - j2.121$$

$$a = 4 + 3 = 7$$

$$b = -1 + 5 = 4$$

$$c = [3(-1) + 4] \omega_8^0 = 1$$

$$d = [5(-1) + (-1)] \omega_8^1 = 6j$$

$$e = -2 + 5j$$

$$f = (3.535 - j3.535) + (-2.121 - j2.121) = 5.656 - j5.656$$

$$g = [5(-1) + (-2)] \omega_8^0 = -2 - j5$$

$$h = [(-2.121 - j2.121)(-1) + (3.535 - j3.535)] \omega_8^1$$

3rd stage calculation

$$x(0) = 7 + 4 = 11$$

$$x(4) = [4(-1) + 7] \omega_8^0 = 3$$

$$x(2) = 1 + 6j$$

$$x(6) = (6j(-1) + 1) \omega_8^0 = 1 - 6j$$

$$x(1) = -2 + 5j + (0.656 - j1.414) = 3.656 + j3.586$$

$$x(5) = [(3.656 - j1.414)(-1) + (-2 + 5j)] \omega_8^0 = -7.656 + j6.414$$

$$x(3) = (-2 - j5) +$$

$$x(7) = (-1.414 - j5.656)$$

$$\therefore x(k) = \{11, 3, 1 + 6j, 1 - 6j\}$$

solⁿ:- Find the 8-point DFT of the sequence $\{1, 1, 2, 4, 3, 1, 2, 1\}$ using radix-2 decimation in time domain.

Solⁿ:- Here, $x(n) = \{1, 1, 2, 4, 3, 1, 2, 1\}$

using radix-2 decimation in time algorithm

1st stage calculation

$$A = 1 + 3 \cdot \omega_8^0 = 4$$

$$B = 3 \cdot \omega_8^0(-1) + 1 = -2$$

$$C = 2 + 2 \cdot \omega_8^0 = 4$$

$$D = 2 + 2 \cdot \omega_8^0(-1) = 0$$

$$E = 1 + 1 \cdot \omega_8^0 = 2$$

$$F = 1 \cdot \omega_8^0(1) + 1 = 0$$

$$G = 4 + 1 \cdot \omega_8^0 = 5$$

$$H = 1 \cdot \omega_8^0(-1) + 4 = 3$$

2nd stage calculation

$$a = 4 + 4 \cdot \omega_8^0 = 8$$

$$b = -2 + 0 \cdot \omega_8^0 = -2$$

$$c = (1 \cdot \omega_8^0(-1) + 4) = 0$$

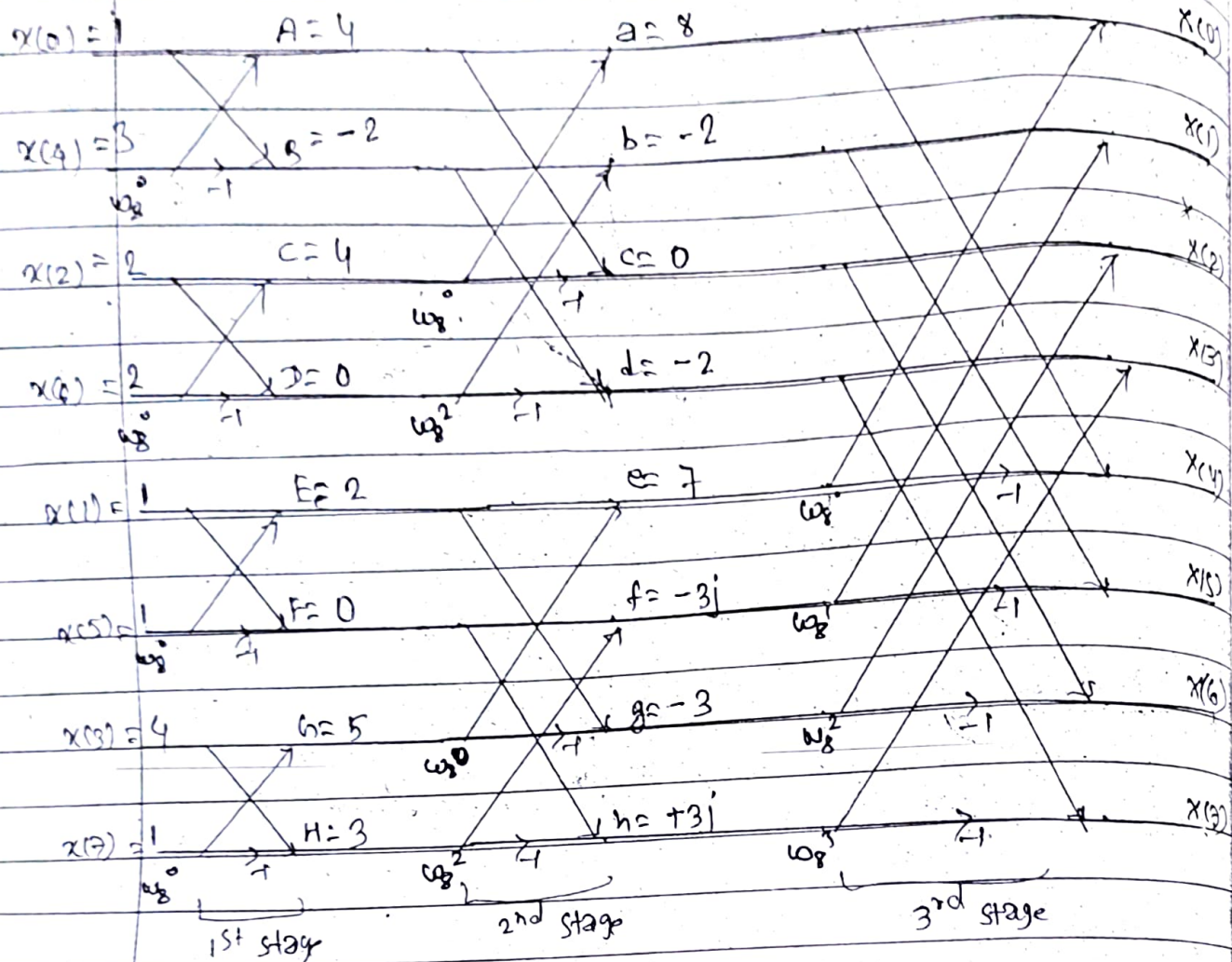
$$d = 0 \cdot \omega_8^0(-1) + (-2) = -2$$

$$e = 2 + 5 \cdot \omega_8^0 = 7$$

$$f = 0 + 3 \cdot \omega_8^0 = -3j$$

$$g = 5 \cdot \omega_8^0(1) + 9 = -3$$

$$h = 3 \cdot \omega_8^0(-1) + 0 = 3j$$



3rd stage calculation

$$X(0) = 8 + 7 \cdot w_8^0 = 15$$

$$X(1) = -2 + 7(-3j) \cdot w_8^1 = -4.121 - j2.121$$

$$X(2) = 0 + (-3j) w_8^2 = 3j$$

$$X(3) = -2 + (3j) w_8^3 = 0.121 - j2.121$$

$$X(4) = 7 \cdot w_8^0 (-1) + 8 = 1$$

$$X(5) = (-3j)(w_8^1)(-1) + (-2) = 0.121 + j2.121$$

$$X(6) = (-3j)(w_8^2)(-1) + 0 = -3j$$

$$X(7) = 3j(w_8^3)(-1) + (-2) = -4.121 + j2.121$$

$$\therefore X(k) = \{15, -4.121 - j2.121, 3j, 0.121 - j2.121, 1, 0.121 + j2.121, -3j, -4.121 + j2.121\}$$

find 8-point DIT-FFT of sequence $x(n) = u(n) + u(n-2) - u(n-6) - u(n-8)$.

Here $x(n) = u(n) + u(n-2) - u(n-6) - u(n-8)$, $n = 0, 1, \dots, 7$

For $n=0$, $x(0) = u(0) + u(2) - u(6) - u(8) = 1$

For $n=1$, $x(1) = u(1) + u(3) - u(5) - u(9) = 1$

For $n=2$, $x(2) = u(2) + u(0) - u(4) - u(6) = 1 + 1 = 2$

For $n=3$, $x(3) = u(3) + u(1) - u(3) - u(5) = 1 + 1 = 2$

For $n=4$, $x(4) = u(4) + u(2) - u(2) - u(4) = 1 + 1 = 2$

For $n=5$, $x(5) = u(5) + u(3) - u(1) - u(3) = 1 + 1 = 2$

For $n=6$, $x(6) = u(6) + u(4) - u(0) - u(2) = 1 + 1 = 2$

For $n=7$, $x(7) = u(7) + u(5) - u(1) - u(3) = 1 + 1 - 1 - 1 = 0$

$\therefore x(n) = \{1, 1, 2, 2, 2, 2, 2, 0\}$

first stage calculation

$A = 1 + 2 \cdot \omega_8^0 = 3$

$B = 2 \cdot \omega_8^1(-1) + 1 = -1$

$C = 2 + 2 \cdot \omega_8^2 = 4$

$D = 2 \cdot \omega_8^3(-1) + 2 = 0$

$E = 1 + 2 \cdot \omega_8^4 = 3$

$F = 2 \cdot \omega_8^5(-1) + 1 = -1$

$G = 2 + 1 \cdot \omega_8^6 = 3$

$H = 1 \cdot \omega_8^7(-1) + 2 = 1$

2nd stage calculation

$a = 3 + 4 \cdot \omega_8^0 = 7$

$b = -1 + 0 \cdot \omega_8^2 = -1$

$c = 4 \cdot \omega_8^3(-1) + 3 = -1$

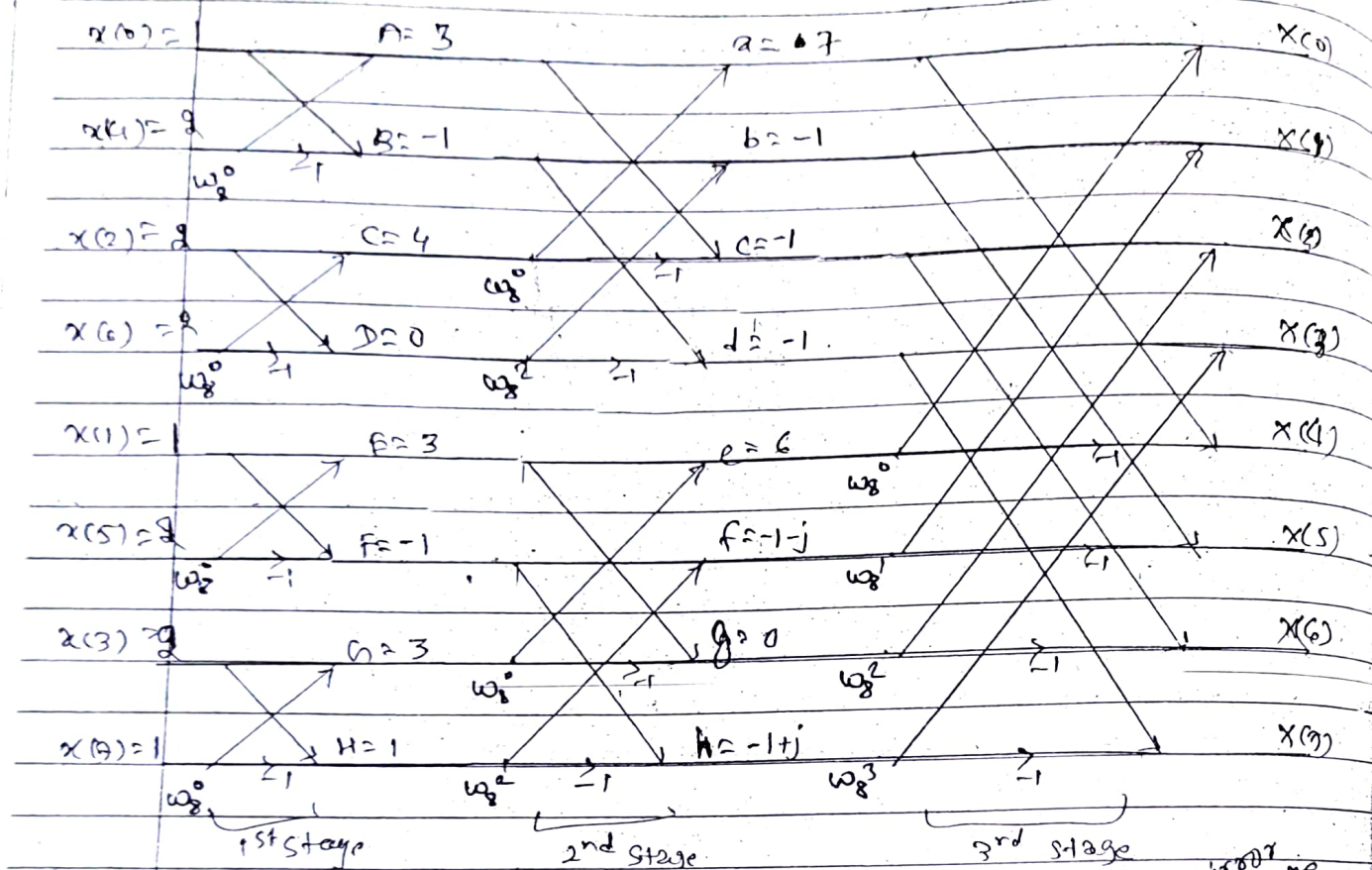
$d = 0 \cdot \omega_8^2(1) + (-1) = -1$

$e = 3 + 3 \cdot \omega_8^4 = 6$

$f = -1 + 1 \cdot \omega_8^6 = -1 - j$

$g = 3 \cdot \omega_8^5(-1) + 3 = 0$

$h = 1 \cdot \omega_8^7(1) + (-1) = -1 + j$



3rd stage calculation

$$X(0) = 7 + 6 \cdot w_8^0 = 13$$

$$X(1) = -1 + (-1-j) w_8^1 = -1 - j = -2.414$$

$$X(2) = -1 + 0 \cdot w_8^2 = -1$$

$$X(3) = -1 + (-1+j) w_8^3 = 0.414$$

$$X(4) = 6 \cdot w_8^4 (-1) + 7 = 1$$

$$X(5) = (-1-j) w_8^5 (-1) + (-1) = 0.414$$

$$X(6) = 0 \cdot w_8^6 (-1) + (-1) = -1$$

$$X(7) = (-1+j) w_8^7 (-1) + (-1) = -1 - j = -2.414$$

$$\therefore X(k) = \{13, -2.414, -1, 0.414, 1, 0.414, -1, -2.414\}$$

$$000 \rightarrow 000 \rightarrow 0$$

$$001 \rightarrow 100 \rightarrow 4$$

$$010 \rightarrow 010 \rightarrow 2$$

$$011 \rightarrow 110 \rightarrow 6$$

$$100 \rightarrow 001 \rightarrow 1$$

$$101 \rightarrow 101 \rightarrow 5$$

$$110 \rightarrow 011 \rightarrow 3$$

$$111 \rightarrow 111 \rightarrow 7$$

mirror image

Find 4-point DFT on a continuous input signal given by $x(t) = \sin 2\pi (1000 t) + \sin(2\pi 2000 t + \frac{3\pi}{4})$ with sampling freq. $f_s = 8 \text{ kHz}$.

here $x(t) = \sin(2\pi 1000 t) + \sin(2\pi 2000 t + \frac{3\pi}{4})$
 $f_s = 8 \text{ kHz}$.

When sampling at $f_s = 8 \text{ kHz}$ i.e. :

$$x(n) = x_2(nT) = x_2(\frac{n}{f_s})$$

$$\therefore x(n) = x(n) = \sin(2\pi 1000 \frac{n}{8000}) + \sin(2\pi 2000 \frac{n}{8000} + \frac{3\pi}{4})$$

$n = 0, 1, 2, 3$

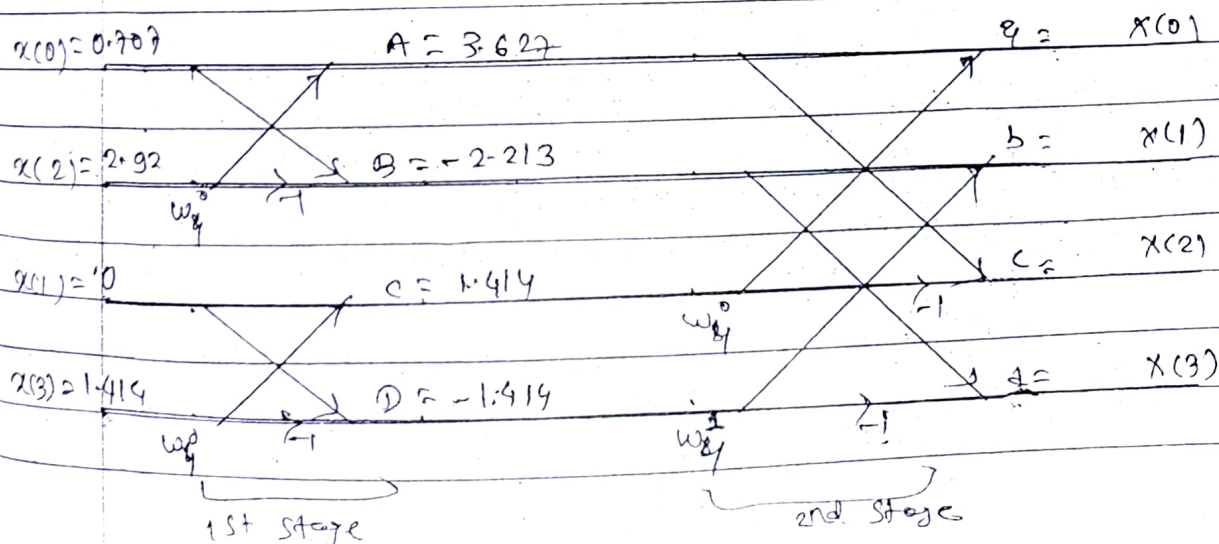
when putting $n = 0, 1, 2, 3$, set by step in $x(n)$, we get

$$x(n) = \{0.707, 0, 0.292, 1.414\}$$

for 8-point: $x(n) = \{0.707, 0, 0.292, 1.414, 0.707, -0.707, -1.707, 0\}$

but we have to compute for 4-point DFT. so,

$$x(n) = \{0.707, 0, 0.292, 1.414\}$$



1st stage calculation

2nd stage calculation

$$A = 0.707 + 2.92 \cdot w_8^0 = 3.627$$

$$B = 2.92 \cdot w_8^1(-1) + 0.707 = -2.213$$

$$C = 0 + 1.414 \cdot w_8^0 = 1.414$$

$$D = 1.414 \cdot w_8^1(-1) + 0 = -1.414$$

$$a = X(0) = 3.627 + 1.414 \cdot w_4^0 = 5.041$$

$$b = X(1) = -2.213 + (-1.414) \cdot w_4^1 = -2.213 + j1.414$$

$$c = X(2) = 1.414 \cdot w_4^0(-1) + 3.627 = 2.313$$

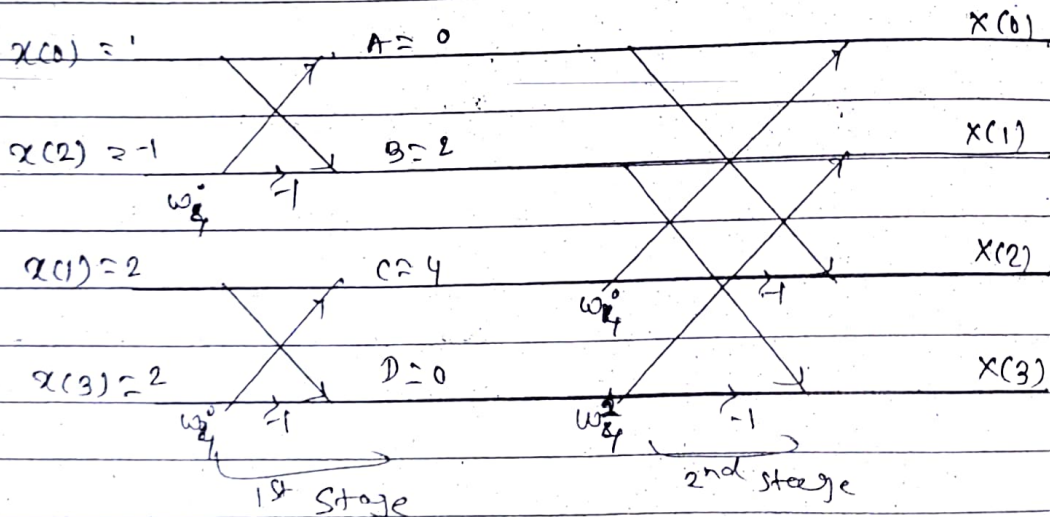
$$d = X(3) = 1.414 \cdot w_4^1(-1) + (-2.213) = -2.213 - j1.414$$

$$\therefore X(K) = \{5.041, -2.213 + j1.414, 2.313, -2.213 - j1.414\}$$

11-Spring

Using DIT FFT radix-2 algorithm; find DFT of $x(n) = \{1, 2, -1, 2\}$

Soln: Here, $x(n) = \{1, 2, -1, 2\}$



1st stage calculation

2nd stage calculation

$$A = 1 + (-1) \cdot w_8^0 = 0$$

$$B = (-1) \cdot w_8^1(-1) + 1 = 2$$

$$C = 2 + 2 \cdot w_8^0 = 4$$

$$D = 2 \cdot w_8^1(-1) + 0 = 0$$

$$X(0) = 0 + 4 \cdot w_4^0 = 4$$

$$X(1) = 2 + 0 \cdot w_4^1 = 2$$

$$X(2) = 4 \cdot w_4^0(-1) + 0 = -4$$

$$X(3) = 0 \cdot w_4^1(-1) + 2 = 2$$

$$\therefore X(K) = \{4, 2, -4, 2\}$$